

Auteur: Abdellah El Moussaoui

Studierichting: Informatica

Oefeningenreeks 2E nr. 4.13

$$\begin{aligned} & \lim_{x \rightarrow \infty} \left(\frac{2x-2}{2x+1} \right)^{2x} \\ &= \lim_{x \rightarrow \infty} \left(\frac{2x+1-3}{2x+1} \right)^{2x} \\ &= \lim_{x \rightarrow \infty} \left(1 + \frac{-3}{2x+1} \right)^{2x} \\ &= \lim_{x \rightarrow \infty} \left(\left(1 + \frac{-3}{2x+1} \right)^{\frac{2x+1}{-3}} \right)^{\frac{-6x}{2x+1}} \\ &= \lim_{x \rightarrow \infty} e^{\frac{-6x}{2x+1}} \\ &= e^{-3} \end{aligned}$$

References